QPCODE: **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION EXAMINATION
V SEMESTER
MG5DSCCFS300-DATABASE MANAGEMENT SYSTEM AND SECURITY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hrs **Maximum Marks: 50 Marks**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Relate data inconsistency to data redundancy in file system data management. [Understand] [CO:1]
2. Identify which data independence deals with changes in physical storage [Understand] [CO:1]
3. Associate entities with real-world objects in an ER diagram. [Understand] [CO:1]
4. Classify the three levels of data abstraction according to their functions. [Understand] [CO:1]
5. Outline the main feature of sequential file organization. [Analyse] [CO:2]
6. Distinguish a candidate key from a non-key attribute. [Analyse] [CO:2]

7. Simplify the purpose of normalization in one statement. [Analyse] [CO:2]

8. Compare Tuple Relational Calculus and Domain Relational Calculus. [Analyse] [CO:2]

9. Show the need for security mechanisms in a database. [Apply] [CO:3]

10. Predict the result of inserting an invalid foreign-key value. [Apply] [CO:3]

11 Compute the valid domain values for a given attribute. [Apply] [CO:3]

12. Classify integrity constraints used in databases. [Apply] [CO:3]

[1x10 = 10]

Part B

Short Essay Type Questions
Answer any **4** questions
Each question carries **5** marks

13. Explain the characteristics of the database approach with suitable examples. [Understand][CO:1]

14. Summarize the main features of ER, relational, and EER models. [Understand][CO:1]

15. Examine the role of keys in the relational data model. [Analyse] [CO:2]

16. Connect transitive dependency with 3NF and illustrate how decomposition removes the dependency. [Analyse] [CO:2]

17. Show how authentication and authorization can be applied to protect a student database containing personal and academic information. [Apply] [CO:3]

18. A database contains three security levels: Confidential, Secret, and Top Secret. Users have corresponding clearance levels. Classify which users can access each level of information under MAC. [Apply] [CO:3]

[5x4 = 20]

Part C

Essay Type Questions
Answer any 2 questions
Each question carries 10 marks

19. Describe database system architecture and explain the different levels of data abstraction, schema, instance, and data independence. [Understand][CO:1]

20. Analyze the following relation: [Analyse][CO:2]

    COURSE(Course_ID, Course_Name, Faculty_ID, Faculty_Name, Department_ID, Department_Name)

    Given appropriate functional dependencies, inspect the relation for normalization problems and illustrate its decomposition into 1NF, 2NF, and 3NF.

21. A database owner wants to give different privileges to the Manager, Accountant, and Clerk. Apply Discretionary Access Control (DAC) by specifying suitable privileges such as SELECT, INSERT, UPDATE, and DELETE. Explain how privileges can be granted and revoked. [Apply] [CO:3]

[10x2 = 20]